

Pharmacodynamic and pharmacokinetic study of oral Curcuma extract in patients with colorectal cancer.

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Curcuma spp. extracts, particularly the dietary polyphenol curcumin, prevent colon cancer in rodents. In view of the sparse information on the pharmacodynamics and pharmacokinetics of curcumin in humans, a dose-escalation pilot study of a novel standardized Curcuma extract in proprietary capsule form was performed at doses between 440 and 2200 mg/day, containing 36-180 mg of curcumin. Fifteen patients with advanced colorectal cancer refractory to standard chemotherapies received Curcuma extract daily for up to 4 months. Activity of glutathione S-transferase and levels of a DNA adduct (M(1)G) formed by malondialdehyde, a product of lipid peroxidation and prostaglandin biosynthesis, were measured in patients' blood cells. Oral Curcuma extract was well tolerated, and dose-limiting toxicity was not observed. Neither curcumin nor its metabolites were detected in blood or urine, but curcumin was recovered from feces. Curcumin sulfate was identified in the feces of one patient. Ingestion of 440 mg of Curcuma extract for 29 days was accompanied by a 59% decrease in lymphocytic glutathione S-transferase activity. At higher dose levels, this effect was not observed. Leukocytic M(1)G levels were constant within each patient and unaffected by treatment. Radiologically stable disease was demonstrated in five patients for 2-4 months of treatment. The results suggest that (a) Curcuma extract can be administered safely to patients at doses of up to 2.2 g daily, equivalent to 180 mg of curcumin; (b) curcumin has low oral bioavailability in humans and may undergo intestinal metabolism; and (c) larger clinical trials of Curcuma extract are merited.